VPN-LAN Service Access — Integration Status

Revision: 1 **Last modified:** 2026-07-01T16:35:00Z **Status:** In progress — Phase 0 (bridge scaffold) + Phase 8 (Miracast verdict) + Phase 9 (docs) DONE; Phases 2/3/4 protocol tests authored + honest-SKIP-proven (live round-trips operator-gated); Phases 5/6/7/10/11 in progress. Every autonomous claim below cites a real `qa-results/` artefact or a committed file (§11.4.6); every live-round-trip path honestly SKIPs until the operator connects the svord bridge (§11.4.3). **Authority:** Inherits `constitution/Constitution.md` per §11.4.35. §11.4.45 integration-status doc for the VPN-LAN service-access feature workstream (`feature/vpn-aware-dynamic-routing`). **Companion:** summary `Status_Summary.md` · plan `PLAN.md` · Miracast verdict `miracast_verdict.md` · operator guide `../../guides/vpn_lan_bridge_setup.md`

Operator-blocked / pending items (read first — §11.4.45 O(1) surface)

Item	Why blocked / pending	Unblock condition
Live protocol round-trips (SMB/NFS/FTP/SFTP/WebDAV/email/Cast/ADB)	Need the live svord bridge connection — secrets + Mullvad WG + root/sudo on the bridge side (§11.4.21 / §11.4.10). The autonomous slate proves the honest-SKIP path + the anti-bluff gate now; the live round-trip evidence exists only when the bridge is genuinely up.	Operator copies `.env.example`→`.env`, points it at `svord_toolkit`, brings the VPN up; then each test emits real captured round-trip evidence.
Phase 5 discovery reflector — remote deploy	Deploying an Avahi/SSDP reflector changes a remote host (§11.4.122) — requires an interactive keep/deploy decision. Local reflector config + a	Operator authorizes the remote-side reflector deployment.

Item	Why blocked / pending	Unblock condition
	local-stub discovery test are autonomous.	
Phase 7 ADB flash (`usbip` fastboot) on real hardware	Flashing a real device is high-blast-radius (§11.4.133) — operator-gated. `adb connect/shell/getprop` over routed 5555 is autonomous once the bridge is up.	Operator authorizes the specific device flash.

Phase status matrix (captured-evidence-driven — §11.4.5/§11.4.69)

Phase	Scope	Status	Evidence / commit
0	env-var svord bridge scaffold + svord-doctor preflight	PASS	`d781002`; svord-doctor 3-verdict discrimination (UP/SKIP/MISCONFIGURED) + standing-suite `test_vpn_lan_bridge` GREEN (honest SKIP + §11.4.115 teeth PASS, `qa-results/suite/run_20260701T160156Z.log` 71/64/7/0) `5c28f56`
1	L3 routed gateway + SSRF allowlist reconciliation	PENDING	Design in `PLAN.md` §4; security-critical, conductor-owned; local-stub SSRF-teeth logic to land before any live 10/8 carve-out (gated on bridge up)
2	SMB/CIFS/NMB + NFS round-trip	AUTHORED (SKIP-proven)	`182e80a` `tests/vpn_lan/smb_nfs_roundtrip.sh`; bridge-down SKIP + exit 0, PASS_lines=0 (no fake PASS); live sha256 round-trip operator-gated
3	FTP/FTPS/SFTP + WebDAV (WebDAV via existing Squid)	AUTHORED (SKIP-proven)	`182e80a` `tests/vpn_lan/ftp_sftp_webdav.sh`; wrong-answer⇒FAIL (WebDAV non-207 / SFTP sha-mismatch),

Phase	Scope	Status	Evidence / commit
			unreachable⇒SKIP; live operator-gated
4	IMAP/IMAPS/ SMTP-submission/ POP3S + open-relay guard	AUTHORED (SKIP-proven)	`2f31460` `tests/vpn_lan/email_roundtrip.sh`; open-relay negative test (external-RCPT-accepted⇒FAIL); creds via stdin never argv (§11.4.10); live operator-gated
5	mDNS/SSDP/ WS-Discovery/ DNS-SD reflector	IN PROGRESS	Remote deploy operator-gated (§11.4.122); reflector design + local-stub discovery test in flight
6	Chromecast / DIAL (reflect discovery + route control)	IN PROGRESS	`tests/vpn_lan/chromecast_dial.sh` authoring in flight; control 8008/8009 routes, discovery via Phase-5 reflector
7	ADB over VPN (access/debug/ connect/flash)	IN PROGRESS	`tests/vpn_lan/adb_over_vpn.sh` authoring in flight; routed 5555 + adb-server; flash via `usbip` (USB-bound, operator-gated §11.4.133)
8	Miracast structural-impossibility verdict	PASS (Won't-fix)	`12faf12` `miracast_verdict.md` (§11.4.112); cited Wi-Fi-Alliance evidence; Google Cast as the routable alternative; no fake traversal test
9	Documentation (operator bridge-setup guide)	PASS	`a5e5616` `../../guides/vpn_lan_bridge_setup.md` (routing map + verdict table + 12-row protocol matrix + FAQ); HTML+PDF §11.4.168 leak-clean
10	Containerization via containers submodule (§11.4.76)	PENDING	Reflector + adb-server to boot on-demand via `submodules/containers` (rootless §11.4.161); depends on Phase 5/6/7 design

Phase	Scope	Status	Evidence / commit
11	§11.4.169 full test-type coverage + Challenges + HelixQA	IN PROGRESS	Challenge script + HelixQA `vpn_lan.yaml` bank authoring in flight; HelixQA-run honestly blocked (6 un-vendored siblings, §11.4.3)

Honest boundary (§11.4.6)

The feature's design + decoupled bridge + anti-bluff test scaffolding are proven now: the Phase-0 bridge is GREEN in the standing suite with a genuine up↔down discrimination (§11.4.115 teeth), and Phases 2/3/4 tests refuse to fake a PASS when the bridge is down (bridge-down ⇒ honest SKIP, exit 0, zero `^PASS:` lines) — the anti-bluff gate is the value delivered autonomously. Miracast is honestly classified structurally-impossible (§11.4.112) with Google Cast as the routable alternative. The **live** protocol round-trip evidence (real bytes, sha256, mailbox content, device serials) exists only once the operator connects the svord bridge (secrets + Mullvad + root/sudo) — until then those paths honestly SKIP (§11.4.3), never a fake PASS. No phase is "done" for the live capability until its runtime signature verifies with captured evidence on a genuinely-up bridge (§11.4.108) and it crosses the §11.4.169 test-type matrix. This doc is §11.4.45-synced + §11.4.65-exported; it does not substitute for the §11.4.40 full-suite retest before any release tag.